

RAMDEOBABA UNIVERSITY



Department of Computer Science and Engineering

PROJECT SYNOPSIS

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AGENTIC AI: AUTONOMOUS INSURANCE CLAIMS TRIAGE & FRAUD ASSISTANT

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1. Background and Motivation

Insurance fraud is estimated to cost the global insurance industry tens of billions of dollars every year, with automobile insurance among the most heavily targeted lines because of the volume and diversity of claims filed. As claim volumes grow, insurers increasingly rely on manual review by human adjusters, a process that is slow, inconsistent across reviewers, and difficult to scale. Most existing fraud-detection systems are either static rule engines that fail to adapt to new fraud patterns, or standalone machine learning classifiers that output a single probability score without supporting context or explanation. This leaves the adjuster to manually gather evidence — comparable historical claims, policy history, and documentation checks — before a decision can be reached, which slows down genuine claims and delays investigation of fraudulent ones.

Recent advances in Large Language Model (LLM)-based agentic systems make it possible to move beyond a single predictive score toward a system that reasons through a claim the way a human adjuster would: extracting relevant details, retrieving similar past cases, applying policy checks, and producing a transparent, explainable recommendation. Major banks and insurers are already exploring this direction, pairing predictive ML models with agentic workflows to reduce manual review time while keeping a human in the loop for the final decision. This project is motivated by that industry direction — to design and build a working prototype of an Agentic AI system for automobile insurance claims that combines a trained fraud-scoring model with an autonomous reasoning agent, giving hands-on exposure to the emerging intersection of applied machine learning and agentic AI system design.

2. Problem Statement

Traditional fraud-detection systems in insurance rely on either static rule engines or standalone ML classifiers that output only a bare probability score, leaving the adjuster to manually gather supporting context — similar past claims, policy history, and documentation gaps — before making a decision. This creates two compounding problems: a high manual workload, since adjusters spend more time gathering context than exercising judgement, slowing down genuine claims and delaying fraud investigation; and low explainability, since a bare fraud score gives no reasoning, making it difficult for adjusters and regulators to trust or audit the decision. The problem

this project addresses is the design and development of an agentic AI system that autonomously performs the multi-step reasoning a human adjuster would otherwise perform manually — extract, validate, score, compare, and recommend — while producing a transparent, auditable rationale for every decision, with the human adjuster retaining final authority to approve or override the recommendation.

3. Objectives

- Build an ML/DL model that predicts the probability that a given insurance claim is fraudulent.
- Design an autonomous agent (LLM-based orchestrator) that plans and executes a multi-step triage workflow using tools rather than a single fixed inference call.
- Implement a similar-claims retrieval mechanism so the agent can compare a new claim against historical patterns.
- Generate a structured, human-readable decision with rationale for each claim (approve / flag / request more information).
- Build a lightweight human-in-the-loop dashboard so an adjuster can review, override, and provide feedback.
- Close the feedback loop so adjuster decisions can be used to retrain and improve the underlying ML/DL model over time.

4. Literature Review and Research Gap

Existing literature on insurance fraud detection is dominated by supervised machine learning approaches. Khalil et al. [1] address class imbalance and missing values on real auto-insurance datasets using resampling and imputation strategies, while Urunkar et al. [3] apply a Random Forest classifier to reduce manual investigation effort. Dhieb et al. [2] go further by proposing a full AI-driven pipeline for scoring and risk measurement, and Yankol-Schalck [8] surveys hybrid ML-DL architectures that push detection accuracy higher still. Across this body of work, the consistent output is a fraud-probability score or binary label: none of these systems retrieve comparable historical claims, apply policy-based reasoning, or generate an explanation an adjuster or regulator can audit.

Separately, work on LLM-based agents — most notably the ReAct framework of Yao et al. [4], which interleaves reasoning traces with tool calls in a loop — has shown that language models can plan multi-step tasks and use external tools autonomously, but this line of research is largely evaluated on open-domain or general-purpose tasks rather than regulated, high-stakes domains such as insurance.

Research Gap: Existing research mainly focuses on ML-based insurance fraud detection and, more recently, agentic AI for claim processing. However, there is an opportunity to combine fraud scoring, similar-claim retrieval, policy checks, and agentic reasoning into a human-in-the-loop fraud-triage system that provides an explainable and evidence-based recommendation to the adjuster.

Sr. No.	Author(s) & Year	Title / Focus	Methodology / Key Contribution	Limitation / Gap
1	Khalil et al., 2024 [1]	ML-based insurance fraud detection on imbalanced, missing-value data	Compares resampling and imputation strategies with multiple ML classifiers on two auto-insurance datasets	Improves score accuracy only; no reasoning, retrieval, or explanation is generated for the adjuster
2	Dhieb et al., 2020 [2]	AI-driven architecture for automated insurance fraud detection and risk measurement	Proposes a secure end-to-end ML pipeline for scoring and risk measurement in insurance workflows	Architecture is a fixed pipeline; it does not use an autonomous agent that plans steps or calls tools dynamically
3	Urunkar et al., 2022 [3]	Fraud detection and analysis for insurance claims using ML	Uses a Random Forest classifier on claim records to flag fraud and reduce manual investigation time	Output is a single classification label with no supporting rationale or similar-case comparison
4	Yao et al., 2022 [4]	ReAct: synergizing reasoning and acting in language models	Introduces interleaved reasoning-and-action traces enabling an LLM to plan, call tools, and revise its actions	General-purpose framework; not applied or evaluated in a regulated, high-stakes domain such as insurance
5	Yankol-Schalck, 2026 [8]	Auto insurance fraud detection: ML and DL applications	Surveys and benchmarks ML and DL techniques, including hybrid ML-DL frameworks, for auto-insurance fraud	Focuses on model accuracy improvements; does not address human-in-the-loop review or decision explainability

Sr. No.	Author(s) & Year	Title / Focus	Methodology / Key Contribution	Limitation / Gap
6	Devender et al., 2026 [9]	Agentic AI for insurance claim processing	Multi-agent system using LLaMA 3.1 for fraud, policy, eligibility, and decision analysis	Lacks ML/DL scoring, similar-claim retrieval, and human-in-the-loop feedback

5. Proposed Methodology / Plan of Work

The system is organised into six phases that move a claim from raw intake to a human-approved decision, with a feedback loop back into the ML model.

Phase 1 — Ingestion & Information Extraction

The system begins by collecting incoming insurance claim data from available sources. This may include structured information such as policy details, claim amount, incident date, location, claimant information, and claim type, along with optional unstructured information such as the claimant’s description of the incident. The collected information is parsed, relevant entities and attributes are extracted, and all data is transformed into a standardized common schema. This ensures that claims received in different formats can be processed consistently by the subsequent stages.

Phase 2 — Data Validation & Enrichment

The standardized claim undergoes validation to identify missing, incomplete, or inconsistent information. Basic business-rule checks are performed, such as verifying whether the incident date falls within the policy coverage period and whether the claim details are logically consistent. After validation, the system performs feature engineering by deriving relevant attributes from the available claim data. Additional contextual information, where available, can also be incorporated to create a richer representation of the claim for fraud detection.

Phase 3 — ML/DL Fraud Probability Scoring

The validated and enriched claim is passed to a trained Machine Learning/Deep Learning classifier. The model analyzes patterns across multiple claim attributes to determine the likelihood that the claim may be fraudulent. Instead of producing only a binary result, the model generates a fraud probability score, which provides a measurable indication of the claim's risk level. This score becomes one of the key inputs for the subsequent agentic reasoning and decision-making process.

Phase 4 — Agentic Reasoning Core

The Agentic Reasoning Core acts as the central intelligent component of the proposed system. An LLM-based orchestrator analyzes the claim and dynamically determines what additional information or verification steps are required. Unlike a fixed rule-based workflow, the agent can select different tools depending on the characteristics of the claim. These tools may include similar-claim retrieval using a vector database, policy verification, claimant history checks, and other relevant information sources. The retrieved evidence is then interpreted and organized to provide contextual reasoning around the ML-generated fraud score.

Phase 5 — Decision Synthesis

The system combines the ML/DL fraud probability score, validation results, retrieved similar claims, policy information, historical information, and the reasoning generated by the agent. Based on this combined evidence, the system produces a structured recommendation. The possible outcomes are Approve, Flag for Investigation, or Request More Information. Along with the recommendation, the system generates a human-readable rationale explaining the major factors and evidence that contributed to the decision, improving transparency and interpretability for the claims adjuster.

Phase 6 — Human Adjuster Review & Feedback

The final recommendation is presented to a human claims adjuster through a lightweight dashboard. The adjuster can review the fraud score, supporting evidence, agent-generated rationale, and recommended action before making the final decision. The adjuster can approve the recommendation, modify it, or override it when human judgment indicates a different outcome. The final decision and reviewer feedback are securely logged and used as feedback data for future

model evaluation and retraining, enabling the system to continuously improve its fraud-detection performance.

Overall Methodology

The proposed methodology follows a human-in-the-loop, ML + agentic AI architecture, where the ML/DL model provides quantitative fraud-risk estimation while the agentic layer performs dynamic evidence gathering and reasoning. Keeping these components modular and separately evaluable allows the ML model and agentic reasoning system to be independently developed, tested, and validated before their integration into the complete insurance claim fraud detection system.

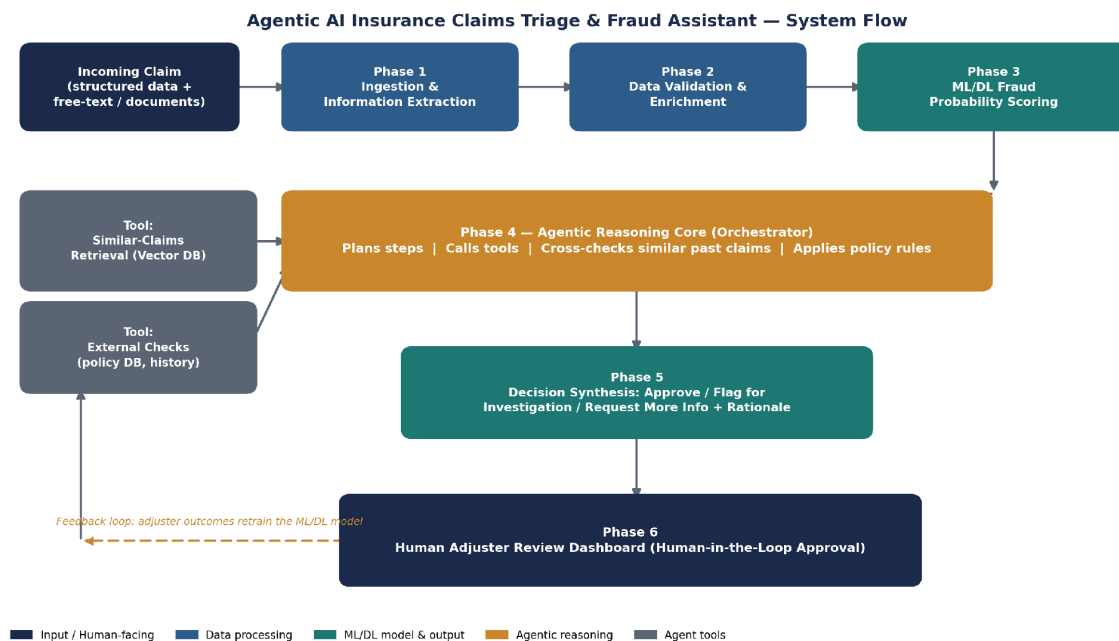


Fig. 1. System workflow of the proposed Agentic AI claims triage pipeline.

6. Technology, Tools and Platforms

1. Programming

- Python – Core programming language

2. ML/DL

- Scikit-learn, XGBoost – Fraud detection models
- TensorFlow/PyTorch – Neural-network variant

3. Agentic AI

- LangChain/LangGraph or CrewAI – Agent orchestration
- FAISS/ChromaDB – Similar-claim vector retrieval

4. LLM

- OpenAI API / Anthropic API / Open-source LLM – Reasoning and decision support

5. Dashboard

- Streamlit or Flask – Human adjuster interface

6. Dataset

- Vehicle Claim Fraud Detection: ~15,420 records, 33 fields
- Auto Insurance Claims Data: Used as the retrieval knowledge base

7. Expected Outcomes, Deliverables and Functional Specifications

The project will deliver a cleaned and documented dataset with an EDA report; a trained ML/DL fraud-scoring model evaluated on accuracy, precision, recall, F1, and AUC-ROC; a functional agentic pipeline comprising an orchestrator and callable tools for fraud scoring, similar-claims retrieval, and policy/history checks; and a working dashboard where an adjuster can submit a claim, view the agent's decision and rationale, and approve or override it. The end-to-end pipeline is expected to run on held-out claims without manual intervention, producing a reviewable recommendation for each, along with a final report and a presentation/demo video for evaluation.

8. Project Scope

In scope: structured claims data (policy, incident, claim-amount fields) as the primary input, with optional free-text claim description handling; a single insurance line (auto/vehicle insurance) to keep the dataset and rules tractable for a student project timeline; an agent built on an open-source or free-tier LLM API with two to three callable tools (fraud model, retrieval, basic policy checks); and a simple, functional — not production-grade — review dashboard.

Out of scope: real-time integration with live insurer core systems or actual customer data; full regulatory/compliance certification (explainability is addressed at a project level, not an audited production level); multi-line insurance support such as health, life, or property, which is listed as future work; and image-based damage assessment from claim photos, which is at most an optional stretch goal.

9. Project Timeline

The project is planned over a nine-week cycle. The first two weeks cover domain study and dataset preparation, followed by two weeks of ML/DL model training and tuning. The agentic orchestrator and its tools are built and integrated into an end-to-end pipeline over the following three weeks, after which the review dashboard is layered on top. The final week is reserved for testing, evaluation, and documentation.

Phase	Duration	Milestone / Deliverable
Literature review & requirement analysis	Week 1 – 2	Finalised dataset, fraud-labelling scheme, and feature list
Design & methodology finalisation	Week 3 – 4	Trained baseline & advanced ML/DL fraud-scoring models with evaluation report
Implementation / development	Week 5 – 7	Working agent orchestrator, retrieval + policy-check tools, and end-to-end pipeline
Testing & evaluation	Week 7 – 8	Adjuster review dashboard integrated; pipeline validated on held-out claims
Documentation & final submission	Week 8 – 9	Final report, presentation deck and demo video prepared for evaluation

10. References

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